We have seen previously that

$$\frac{d}{dx}e^x = e^x.$$

We now find the derivative of $\ln(x)$. It is then a simple matter to find formulas for the derivatives of exponentials and logs of any base.

1.) Fill in the blank: $e^{\ln(x)} =$

2.) Assuming $\frac{d}{dx}\ln(x)$ exists (we will prove existence later), differentiate both sides of your equation above using implicit differentiation to find $\frac{d}{dx}\ln(x)$.

♢ **Check your answer with your instructor before moving on.**

3.) Compute the derivative of each of the following functions:

(a) $y = x\ln(x)$

(b) $y = \ln(x^2)$

(c) $y = \ln[(6x^4+3)^5]$

4.) Consider now the function $f(x) = a^x$, where $a > 0$.

(a) Use log and exponent properties to show that a^x can be written as $a^x = e^{x\ln(a)}$.

(b) Calculate $\frac{d}{dx}a^x$ by differentiating $e^{x\ln(a)}$. What happens when $a = e$?

♢ **Check your answer with your instructor before moving on.**

5.) Differentiate each of the following functions:

(a) $y = \dfrac{10^x}{x}$

(b) $y = 4^{\sec(x)}$

(c) $9^{x^3\ln(x)}$

6.) Recall that if $a > 0$ and $a \neq 1$, then a^x is invertible with inverse $\log_a(x)$.

7.) Use the change of base formula $\log_a(x) = \frac{\ln(x)}{\ln(a)}$ to calculate $\frac{d}{dx}\log_a(x)$.

♢ **Check your answer with your instructor before moving on.**

8.) Differentiate each of the following functions:

(a) $y = \log_2(x)$

(b) $y = \log_{10}(2+\sin(x))$

(c) $y = \csc(x)\log_3(\pi^x)$

9.) A colony of mosquitoes has an initial population of 1000. After t days, the population is given by $A(t) = 1000e^{0.3t}$. What is the rate of change of the population after 4 days? Make sure to include units.

10.) Find the slope of the tangent line to the graph of $y = \log_2(3x+1)$ at the point $(1,2)$.

11.) Find $f'(x)$ if $f(x) = \ln\left(\frac{x^2\sin(x)}{2x+1}\right)$. (*Hint*: Use properties of logarithms.)

12.) Find $\frac{d}{dx}\ln(|x|)$.

♢ **Check yours answers with your instructor.**